Forecasting Sampled Net-Position Peaks: When Do Conditional Quantiles Beat Historical Rules?

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Abstract

Dealers monitor the largest absolute net position reached during a holding period. We ask whether conditional quantile methods forecast this peak better than historical rules. We compare hourly FX and weekly New York Fed dealer positions, matching training histories and update schedules. Historical rules are hard to beat. In FX, regression lowers loss against its matched rule by 14.8%, and by 16.6% in an earlier 2015 period not previously used for evaluation in this study. Its loss against the short-history rolling rule rises 3.2% in that earlier period. On weekly positions, Treasury-bill loss rises 50.8%; stronger regularization still leaves 34 of 48 cells across 12 further series behind historical headroom. Scores and breaches also diverge: the main FX gain accompanies breaches rising from 332 to 376. Peak-position comparisons should therefore report both scores and breaches. The evidence is retrospective, and the original FX evaluation period was inspected before the conditional methods were specified.

Keywords: quantile forecasting; dealer positions; sampled path maxima; forecast evaluation.

1 Introduction

Dealer inventories connect customer order imbalances to risk limits and balance-sheet capacity. Inventory affects optimal dealer pricing and observed position adjustment (Ho and Stoll, 1981; Lyons, 1995); dealer capacity also matters for Treasury intermediation (Duffie et al., 2023). A position can become large during a holding interval and then return toward its starting value. Monitoring its largest recorded magnitude therefore asks a different question from predicting the endpoint.

We ask when conditional quantile methods forecast a sampled absolute position peak better than simple historical rules. The data comprise an hourly FX market-maker record and weekly Treasury-bill positions. An upper quantile of the peak supplies a monitoring report for future recorded positions. Across these records, historical rules are hard to beat.

Historical headroom and translated-path rules compete with ridge quantile regression and nearest-neighbour quantiles. Matched comparisons use the same completed responses and update schedules. The two pre-specified comparisons pair FX regression with short-history rolling headroom and Treasury regression with long-history frozen headroom.

The paper makes three empirical points. First, the historical benchmark matters for FX forecasts. Regression lowers matched loss by 14.8% in the original period and 16.6% in the earlier 2015 period. It beats the short-history rolling rule only in the original period (Tables 3 and 10).

Second, weekly regression struggles against historical headroom. The Treasury-bill main comparison loses 50.8%. Across 12 further weekly series, expanded-menu regression still loses in 34 of 48 cells (Figure 3). Conservative one-standard-error selection also leaves large weekly losses (Section 7).

Third, lower pinball loss need not mean fewer breaches. The main FX gain accompanies 376 breaches against 332 for its rolling reference. Neither main comparison dominates across consistent quantile scores (Table 8; Appendix A.6). All methods are standard; the contribution is this evidence for the sampled peak, extending the endpoint question studied by Wisniewski et al. (2020).

The paper develops the target and methods before comparing forecast performance. It then examines regularization, error amounts and breaches. Mathematical arguments and implementation details follow in the appendices.

2 Related literature

Inventory research provides the economic motivation for position monitoring. Ho and Stoll (1981) model dealer quotes under inventory uncertainty, and Lyons (1995) studies inventory and information in FX trading. Duffie et al. (2023) connects Treasury-market functioning to dealer capacity.

Financial path extrema and liquidity buffers provide the closest target precedents. MaxVaR measures the worst intermediate portfolio-value loss (Boudoukh et al., 2004). Petrunia et al. (2019) studies the maximum end-of-day net debit across payment-system participants, and van den Hoogen (2017) models a one-sided maximum cumulative payment deficit. Arjani et al. (2020) forecasts a payment-demand/resource ratio. Table 1 distinguishes these targets from the absolute peak of a single recorded position series.

Conditional quantile forecasting combines regression with consistent scoring rules. Regression quantiles minimize asymmetric absolute loss (Koenker and Bassett, 1978); CAViaR models time-varying conditional quantiles (Engle and Manganelli, 2004). Volatility-updated historical simulation is an established alternative for financial scenario construction (Hull and White, 1998; Barone-Adesi et al., 1999). Wisniewski et al. (2020) already applies conditional methods and lag/calendar predictors to endpoint intervals in the FX source family. Gneiting (2011) formalizes consistent quantile scores, and Ehm et al. (2016) represents their comparison through elementary scores. These methods give a common basis for forecasting and scoring the peak response.

Table 1: Related studies: targets, data, methods and evaluation evidence.

Study	Target	Data	Method	Evaluation
Boudoukh et al. (2004)	Worst intermediate portfolio-value loss	Modeled portfolio paths	First-passage formula; discrete simulation	Model-implied MaxVaR/VaR ratios; discrete-monitoring simulation
Petrunia et al. (2019)	Largest daily endpoint debit across ACSS participants	ACSS settlement balances	GEV blocks; GPD exceedances	Fitted-versus-empirical quantiles; monthly-updated collateral shortfalls
van den Hoogen (2017)	Maximum cumulative negative payment balance	Two ABN AMRO transaction accounts	ACD/UHF-GARCH; Monte Carlo paths	Coverage and independence tests; application details confidential
Wisniewski et al. (2020)	Next-hour signed-position interval	One market maker's hourly FX net positions	Quantile/conformal intervals; KNN and other regressions	Two endpoint pinball scores; interval and coverage diagnostics
This paper	Two-sided absolute sampled peak quantile	Hourly FX; weekly dealer-position series	Historical rules; QR/KNN on matched histories and updates	Pinball; paired DM; U/V, breaches and elementary scores

Van den Hoogen (2017) specifies coverage and independence tests in Section 5.4; detailed application results in Section 5.5 are confidential, while Chapter 7 provides public out-of-sample summaries.

The comparison adds three empirical findings for a common sampled position peak. Matched regression gains persist in FX but change sign in Treasury (Tables 3–4). The earlier FX gain persists against matched history, while the gain over rolling history disappears (Table 10). Wider penalties reduce the weekly panel's median excess loss from 92.6% to 9.2%, yet conservative selection still leaves large losses (Figure 3; Section 7). Finally, the main FX score gain accompanies more breaches, and both main Murphy curves cross (Table 8; Appendix A.6). These results connect the peak target to benchmark choice, regularization and risk monitoring.

3 Data and target

The FX record contains hourly client-flow net positions for one market maker across five liquid currency pairs, as described by Wisniewski et al. (2020). We use the hourly member of the Net Position Time Series Dataset, version two (davidlindsay1979, n.d.). It contains 18,576 observations. Adjacent observations are usually an hour apart, with weekend-sized gaps retained in their recorded order. The signed scalar is denominated in source-reported USD. The record belongs to the source family of Wisniewski et al. (2020).

The Treasury input is New York Fed Primary Dealer Statistics series PDPOSGS-B, the aggregate Treasury-bill long position less the short position (New York Fed, 2026). The weekly Wednesday-close series contains 706 observations, denominated in USD millions. FR 2004A instructions describe market/fair-value positions on a trade-date basis, including eligible forward and when-issued outright positions, with specified settlement-date treatment for buybacks (Board of Governors, 2013, 2024). Table 2 gives the chronological splits in calendar dates.

Table 2: Source records and chronological validation and evaluation periods, with issue dates and common evaluation origins across horizons.

Source	Full record	Validation	Evaluation	W / d
FX	2014-02-05 to 2017-01-24	2015-10-08 to 2015-12-30	2016-01-01 to 2017-01-23	720 or 2880 / 24
Treasury	2013-04-03 to 2026-10-07	2019-07-24 to 2020-07-15	2021-01-06 to 2026-04-22	52 or 260 / 13

FX has 1,440 validation and 6,625 evaluation origins; Treasury has 52 and 277. Validation responses finish before evaluation starts; H counts recorded observations, including FX gaps.

The target is the largest absolute position over the sampled holding interval. It includes the position known at issue. Let N_t be the signed recorded position and H count subsequent records, with

$$B_t = |N_t|, \qquad P_{t,H} = \max_{0 \le h \le H} |N_{t+h}|, \qquad G_{t,H} = P_{t,H} - B_t \ge 0. \tag{1}$$

Endpoint marginals permit different peak distributions. For example, two future magnitudes can each take values zero or one with equal probability. Perfect positive dependence gives maximum one with probability one half; perfect negative dependence gives maximum one with probability one. Figure 1 illustrates the distinction between an interim peak and the endpoint.

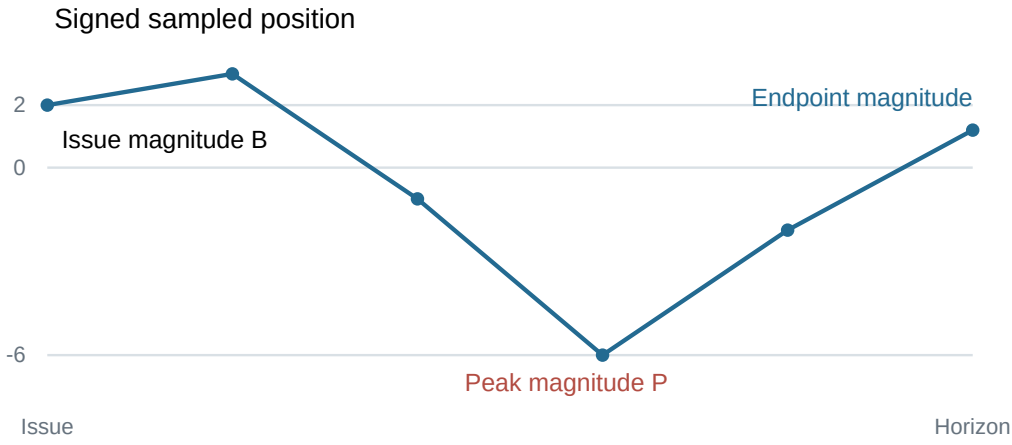


Figure 1: A sampled path separates the issue magnitude, holding-interval peak and endpoint magnitude. The path is artificial.

The information set enforces label completion before fitting. Write $\kappa(t)$ for the latest scheduled refit and W for the number of completed training origins. At segment start t_0 and refit interval d,

$$\kappa(t) = t_0 + d\left\lfloor \frac{t - t_0}{d} \right\rfloor, \qquad \mathcal{T}_{\kappa,H,W} = \{\kappa - H - W + 1, \ldots, \kappa - H\}. \tag{2}$$

Every training response ends by $\kappa(t)$. Fitted transforms and coefficients stay fixed between refits, while current query predictors and B_t update at issue. Earlier evaluation outcomes enter training only after completion. Validation and evaluation restart their schedules at their respective segment starts.

4 Forecasting methods

4.1 Historical reports and matched controls

The historical rules differ in how they anchor completed episodes to the current issue. For each episode i, let $\ell_i = \min_h(N_{i+h} - N_i)$ and $u_i = \max_h(N_{i+h} - N_i)$, including $h = 0$. At query $x = N_t$, its raw translated peak is $T_i(x) = \max\{x + u_i, -x - \ell_i\}$. For a selected set A of size m, Q_j denotes its duplicate-preserving order statistic at $j = \lceil (m+1)\tau \rceil$, with $\tau = .95$. Here F denotes historical headroom, S translated paths and K native peaks. The three reports are

$$\begin{aligned} C_t^F &= B_t + Q_j\{G_{i,H} : i \in A\}, \\ C_t^S &= B_t + Q_j\{T_i(N_t) - B_t : i \in A\}, \\ C_t^K &= \max\{B_t, Q_j\{P_{i,H} : i \in A\}\}. \end{aligned} \tag{3}$$

Rolling controls use the most recent completed episodes at every issue. Frozen controls use $A = \mathcal{T}_{\kappa,H,W}$, matching the conditional fitting pool and update schedule. The query anchor still updates in a frozen control. Appendix A gives the finite-array relationships among these constructions.

4.2 Predictors and training transforms

The original specification contains 33 predictors. They comprise current and lagged signed levels, current magnitude, two past increment root mean square (RMS) measures, calendar sine/cosine pairs and a gap indicator. The level sequence spans lags zero through 24. RMS uses short and long past-only spans, defined by

$$v_L(i) = \left\{ L^{-1} \sum_{q=i-L+1}^{i} (N_q - N_{q-1})^2 \right\}^{1/2}. \tag{4}$$

The spans are daily and weekly record counts in FX, and six and 26 records in Treasury. All transforms use completed training rows. With training mean $\mu_{\kappa j}$ and population standard deviation $\sigma_{\kappa j}$,

$$z_{i,\kappa j} = \frac{x_{ij} - \mu_{\kappa j}}{\sigma_{\kappa j}}, \qquad s_\kappa = \max\left\{ W^{-1} \sum_{i \in \mathcal{T}} G_{i,H},\ 10^{-6} W^{-1} \sum_{i \in \mathcal{T}} |N_i|,\ 10^{-12} \right\}. \tag{5}$$

An exactly constant predictor is centered at its repeated value and assigned scale one. Treasury's calendar and gap coordinates are constant under its weekly sampling. Predictor lookbacks use past observations and can extend beyond the completed-label history. The longer FX RMS reaches before the oldest training origin.

The four-predictor specification uses current signed position, current magnitude and the two past RMS measures. The full specification uses all predictors above. Both select penalties with the same validation rule.

4.3 Quantile regression and nearest neighbours

Ridge quantile regression (QR) fits normalized headroom with an unpenalized intercept. For $\rho_\tau(e) = \tau(e)_+ + (1-\tau)(-e)_+$, it solves

$$\min_{\beta_0,\beta} \frac{1}{W} \sum_{i \in \mathcal{T}} \rho_\tau \left(\frac{G_{i,H}}{s_\kappa} - \beta_0 - z_{i,\kappa}^\mathsf{T} \beta \right) + \frac{\lambda}{2} \|\beta\|_2^2, \tag{6}$$

and issues $C_t^{QR} = B_t + \max\{0, s_\kappa(\widehat{\beta}_0 + z_{t,\kappa}^\mathsf{T}\widehat{\beta})\}$. The original penalty menu is $\{.001, .01, .1\}$. Expanded-menu regression adds the stronger penalties 1 and 10. Projection acts on issued reports; the training objective uses the affine headroom prediction. Appendix A.5 derives the equivalent peak parameterization and its shifted penalty.

As $\lambda \to \infty$, penalized slopes vanish and the unpenalized intercept approaches the set of empirical headroom-quantile minimizers. Write $I_t = B_t + G_{(\lceil W\tau \rceil)}$ for its lower-endpoint reference on the frozen pool. If $W\tau$ is integral, the entire interval between the adjacent order statistics minimizes the intercept-only objective. Thus I is the unique empirical quantile at the shorter Treasury history, while the longer Treasury and both FX histories admit an interval with endpoints I and corrected-rank F (possibly coincident). The different rank conventions can produce different reports.

Nearest-neighbour quantiles (KNN) select standardized predictor rows by a group-balanced Euclidean distance and resolve equal distances by recency. Before computing distance, each lagged-level coordinate is divided by five. Each RMS, hour and weekday coordinate is divided by $\sqrt{2}$. Magnitude and gap retain their standardized scales.

Plain KNN applies the corrected order statistic to selected headrooms. Scaled KNN first divides each selected headroom by a short-run RMS scale and multiplies the resulting quantile by the current scale. The RMS floor is $10^{-6}s_\kappa$. At $k = W$, plain KNN equals frozen F because membership is the full pool.

Standard validation selects the lowest native-unit mean pinball loss. Exact parameter ties favour the larger candidate. Every declared candidate must complete for the cell to be eligible.

4.4 Residual replay

Residual replay adds a fitted conditional mean to historical path construction. At each original refit, ordinary least squares estimates an intercept and slope on unique transitions from the union of completed training windows. Ordered residual blocks are replayed from the current position; Appendix A.4 gives the path identity.

5 Evaluation design

The two pre-specified comparisons use six-record horizons. The FX comparison pairs long-history regression with the short-history rolling headroom rule. The Treasury comparison pairs regression with its long-history frozen headroom rule. We call these the main comparisons below. The original FX evaluation period was inspected before the conditional methods were specified.

Mean pinball loss measures forecast accuracy, while breach frequency records limit exceedances. For report C_t, define the mean underprediction amount U and overprediction amount V by

$$U = \frac{1}{M} \sum_t (P_{t,H} - C_t)_+, \quad V = \frac{1}{M} \sum_t (C_t - P_{t,H})_+, \quad \overline{L} = .95U + .05V. \tag{7}$$

A strict breach occurs when $P_{t,H} > C_t$; equality counts as a nonbreach. Both amounts average over all issues, and headroom is $C_t - B_t$. Scores retain the source units, and percentage changes use the stated reference mean.

Paired loss differences compare unconditional predictive accuracy. For $d_t = L_t^A - L_t^B$, the test statistic is $\bar{d}/\sqrt{\widehat{\Omega}/M}$. We estimate $\widehat{\Omega}$ with Bartlett Newey–West HAC and bandwidth $L = \max\{H, \lfloor 4(M/100)^{2/9} \rfloor\}$ (Diebold and Mariano, 1995; Newey and West, 1987). The original short-horizon FX comparisons use ten lags; the other original cells use the horizon. Twice-bandwidth checks and the Harvey–Leybourne–Newbold correction accompany the original comparisons (Harvey et al., 1997). The fixed-window interpretation follows Giacomini and White (2006).

Apart from the two pre-specified comparisons, all p-values are exploratory and unadjusted unless stated. The main comparisons have unadjusted HAC p-values and a two-comparison Bonferroni sensitivity. Table notes identify the remaining Holm-adjusted comparisons (Holm, 1979).

Coverage diagnostics use the fixed grid $t_0 + j(H + 1)$, starting at evaluation onset. Its target windows share no recorded observations. The Kupiec test uses every retained hit (Kupiec, 1995). Christoffersen independence and conditional-coverage likelihood ratios condition on the first hit and use the subsequent transitions (Christoffersen, 1998). Table 5 reports the resulting tests.

6 Results

6.1 FX: gains depend on history and reference

Long-history regression beats both six-step FX headroom references. Loss falls by 12.7% against rolling headroom and 14.8% against matched frozen headroom. Table 3 places this gain beside the short-history forecasts and other conditional methods.

Table 3: FX forecasts matched by horizon and history, with mean pinball losses in source-reported USD millions.

H	W	Report	Mean loss	Δ_F (%)	Breach (%)	p
6	720	F	0.443	+0.000	5.011	–
6	720	S	0.451	+1.843	3.849	–
6	720	K	0.932	+110.664	7.758	–
6	720	QR	0.433	-2.117	8.015	1.000
6	720	KNN	0.436	-1.394	4.815	1.000
6	720	Scaled KNN	0.466	+5.349	4.060	1.000
6	2880	F	0.453	+0.000	5.434	–
6	2880	S	0.451	-0.418	4.045	–
6	2880	K	0.972	+114.751	8.091	–
6	2880	QR	0.386	-14.767	5.675	< 0.001
6	2880	KNN	0.411	-9.302	4.815	0.016
6	2880	Scaled KNN	0.426	-5.795	4.121	1.000
24	720	F	0.787	+0.000	6.445	–
24	720	S	0.755	-4.078	4.362	–
24	720	K	1.206	+53.135	11.049	–
24	720	QR	1.035	+31.504	13.947	0.066
24	720	KNN	0.836	+6.197	7.592	1.000
24	720	Scaled KNN	0.897	+13.886	6.642	1.000
24	2880	F	0.763	+0.000	5.389	–
24	2880	S	0.759	-0.525	4.408	–
24	2880	K	1.240	+62.440	8.936	–
24	2880	QR	0.735	-3.758	8.015	1.000
24	2880	KNN	0.770	+0.837	6.883	1.000
24	2880	Scaled KNN	0.848	+11.085	5.962	1.000

Relative changes use each report's frozen F; conditional p-values use Holm over the remaining 71 matched conditional-versus-$F/S/K$ contrasts. Dashes mark historical contrasts, self-comparisons or exact full-pool KNN identities; identical losses enter the adjustment with p-value one.

Short-history regression offers little six-step improvement. Its matched gain is 2.12%, with $p = 0.54$. Table 3 contrasts this weak result with the long-history gain.

The short-history 24-step regression loses, even under the expanded penalty menu. Its loss rises by 31.5%, and both menus select $\lambda = 0.01$. The identical selected reports show that menu expansion leaves this failure intact.

The 24-step FX comparison gives a small, imprecise gain. Regression lowers loss by 1.83% against rolling translated paths, with $p = 0.84$. This gain is concentrated in the partial 2017 tail, while the six-step main gain has the same direction in each issue year.

6.2 Treasury: original regression loses to historical headroom

Original Treasury regression loses to matched historical headroom. The main comparison raises loss by 50.8%, with $p = 0.008$. All original regression and scaled-neighbour cells lose to their own frozen headroom references (Table 4).

The Treasury loss increase accompanies more breaches. Their count rises from 28 to 54 in the main comparison. Table 4 shows that the adverse score result extends across both histories and horizons.

Table 4: Treasury forecasts matched by horizon and history, with mean pinball losses in source-reported USD millions.

H	W	Report	Mean loss	Δ_F (%)	Breach (%)	p
6	52	F	2779.054	+0.000	3.971	–
6	52	S	2786.620	+0.272	3.971	–
6	52	K	2574.098	-7.375	10.108	–
6	52	QR	5095.519	+83.354	30.325	0.112
6	52	KNN	2779.054	+0.000	3.971	–
6	52	Scaled KNN	4208.293	+51.429	3.249	0.013
6	260	F	2758.665	+0.000	10.108	–
6	260	S	2707.244	-1.864	10.108	–
6	260	K	2287.758	-17.070	10.469	–
6	260	QR	4159.673	+50.786	19.495	0.008*
6	260	KNN	2663.561	-3.447	10.108	1.000
6	260	Scaled KNN	3127.173	+13.358	8.303	1.000
24	52	F	3715.693	+0.000	11.913	–
24	52	S	3652.379	-1.704	11.913	–
24	52	K	5017.741	+35.042	25.271	–
24	52	QR	12165.574	+227.411	47.653	0.983
24	52	KNN	3715.693	+0.000	11.913	–
24	52	Scaled KNN	5625.667	+51.403	11.552	0.226
24	260	F	2480.306	+0.000	4.332	–
24	260	S	2478.871	-0.058	4.332	–
24	260	K	2301.476	-7.210	24.188	–
24	260	QR	5204.760	+109.843	30.325	1.000
24	260	KNN	2257.963	-8.964	4.693	1.000
24	260	Scaled KNN	4137.264	+66.805	2.888	< 0.001

Relative changes use each report's frozen F; conditional p-values use Holm over the remaining 71 matched conditional-versus-$F/S/K$ contrasts, except the marked unadjusted Treasury main comparison. Dashes mark historical contrasts, self-comparisons or exact full-pool KNN identities; identical losses enter the adjustment with p-value one.

The plain nearest-neighbour forecast reproduces historical headroom at the short Treasury history. Validation selects the full eligible pool at both horizons. At the longer history and horizon, its full-sample gain reverses on the twelve-point nonoverlap grid. Both reports have two breaches on that grid.

Coverage diagnostics reinforce Treasury regression's adverse breach record. The two Treasury grids contain 40 and 12 observations. Table 5 puts the regression tests beside historical headroom and the FX comparison.

Table 5: Six-step coverage diagnostics on the fixed nonoverlap grid.

Report	n	Breaches	Rate (%)	p_{POF}	p_{IND}	p_{CC}
FX QR	947	50	5.280	0.695	0.388	0.636
Treasury QR	40	7	17.500	0.004	0.775	0.014
Treasury F	40	4	10.000	0.199	0.338	0.262

POF uses every retained hit; IND and CC condition on the first hit and use the subsequent transitions in both likelihoods. The p-values use asymptotic equality and independence reference distributions.

6.3 What changes when histories, predictors and penalties change?

Penalty expansion accounts for most of the Treasury score improvement in the ordered comparison. The main regression's excess loss falls from 50.8% to 3.43%. Table 6 holds the original labels and update schedule fixed while expanding the penalty menu.

The two predictor sets give similar Treasury scores under the common menu. Full regression has 1.24% higher loss than four-predictor regression, with $p = 0.685$. Table 7 separates this predictor-set comparison from the final return to the smaller penalty menu.

The FX gain survives the wider penalty menu. Switching from four predictors to the full set lowers long-history loss by 10.9%, with adjusted $p < .001$. Its selected reports equal those from the original full regression (Table 6).

Stronger Treasury shrinkage leaves the breach count above historical headroom. At the main Treasury setting, expanded-menu regression has 32 breaches against 28 for the historical rule. Table 6 also places the ordinary intercept-only quantile beside each regression.

Table 6: Regression loss changes (percent) relative to frozen F under the expanded penalty menu; selected penalties and breaches describe expanded-menu regression.

Source	H	W	λ	Original regression	Four-predictor regression	Expanded-menu regression	I	Breaches
FX	6	720	0.01	-2.12	+1.48	-2.12	-0.15	531
FX	6	2880	0.01	-14.77	-4.29	-14.77	+0.01	376
FX	24	720	0.01	+31.50	+16.51	+31.50	-0.00	924
FX	24	2880	0.001	-3.76	-2.89	-3.76	+0.03	531
Treasury	6	52	10	+83.35	-2.59	-3.24	-1.99	14
Treasury	6	260	1	+50.79	+2.17	+3.43	+3.24	32
Treasury	24	52	10	+227.41	+5.95	+4.34	+6.21	39
Treasury	24	260	10	+109.84	+0.11	-0.19	+2.35	17

Expanded-menu and four-predictor regressions each select from $\{.001, .01, .1, 1, 10\}$; original regression retains its smaller menu. The intercept-only reference I uses $\lceil .95W \rceil$, and F uses $\lceil .95(W+1) \rceil$.

Expanded-menu regression still loses in most additional weekly cells. It loses in 34 of 48 cells. Figure 3 covers 12 further dealer-position series across the same histories and horizons.

Menu expansion nevertheless reduces the typical weekly deterioration substantially. Median excess loss falls from 92.6% to 9.2%. Figure 3 shows both versions against historical headroom for every cell.

Two of the three pre-set panel predictions fail. The number of losing cells falls from 45 to 34. Appendix C gives the fixed prediction thresholds and all three outcomes.

Table 7: Ordered six-step loss changes from historical rules to original regression.

Source	Change	Step (%)	Initial (pp)	p_{DM}	p_{HLN}
FX	Rolling F_{720}→Frozen F	+0.190	+0.190	0.297	0.297
FX	Frozen F_{720}→Frozen F_{2880}	+2.277	+2.281	0.143	0.143
FX	Frozen F→four-predictor regression	-4.289	-4.395	0.010	0.010
FX	four-predictor regression→expanded-menu regression	-10.947	-10.736	$< .001$	$< .001$
FX	expanded-menu regression→original regression	+0.000	+0.000	–	–
Treasury	Frozen F→four-predictor regression	+2.166	+2.166	0.178	0.188
Treasury	four-predictor regression→expanded-menu regression	+1.242	+1.269	0.685	0.691
Treasury	expanded-menu regression→original regression	+45.779	+47.352	0.006	0.007

Step percentages use each preceding loss and compound; initial percentage points share the chain's starting loss and add. The last Treasury step restores the smaller penalty menu, dashes mark identical losses, and paired tests are unadjusted.

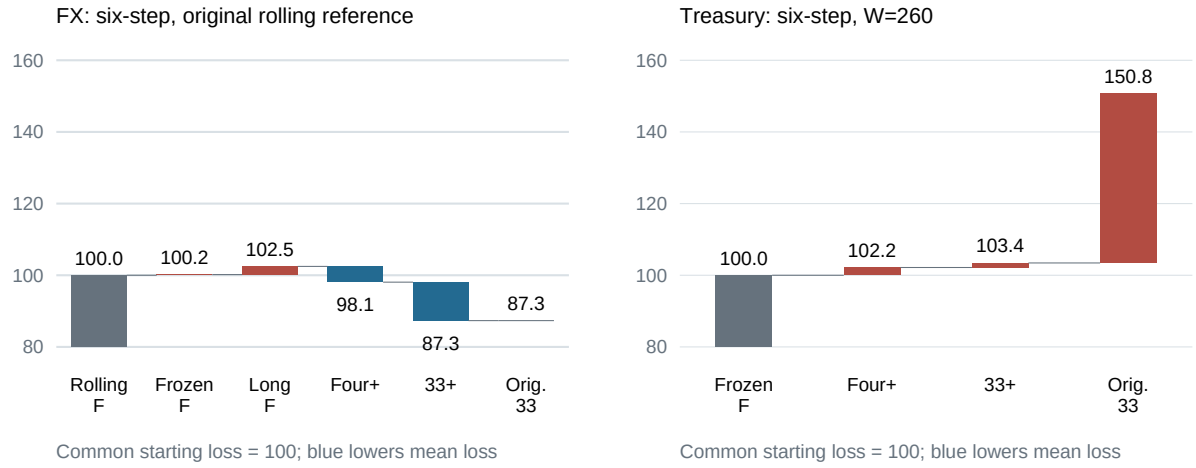


Figure 2: Successive procedure changes in the six-step comparisons. Each increment uses the chain's common starting loss, so the displayed increments sum to the total. Table 7 gives the corresponding paired tests.

7 Robustness and score interpretation

Residual replay improves some scores while increasing breaches. It lowers loss in 5/8 cells against each matched historical reference. Breaches rise against translated paths in every cell and against headroom in 5/8 (Table 9).

Residual replay improves the main Treasury score against both historical references. Loss falls by 20.4% against headroom and 18.9% against translated paths. Its 31 breaches exceed the headroom rule's 28.

Residual replay also beats most Treasury regression reports. It beats expanded-menu regression in 3/4 Treasury cells, including a 23.0% reduction at the main setting. It beats original regression in every Treasury cell; Appendix A.4 gives the path construction.

The historical quantile rank changes both reports and rankings. Corrected headroom uses $\lceil (m+1)\tau \rceil$, while the ordinary empirical quantile uses $\lceil m\tau \rceil$. The ranks differ in the short Treasury pool, and nearest-neighbour headroom equals frozen headroom when both use the full pool and the same rank.

The main FX gain improves both mean error amounts while increasing breaches. The breach count rises from 332 to 376. Table 8 separates the underprediction and overprediction amounts from their frequency.

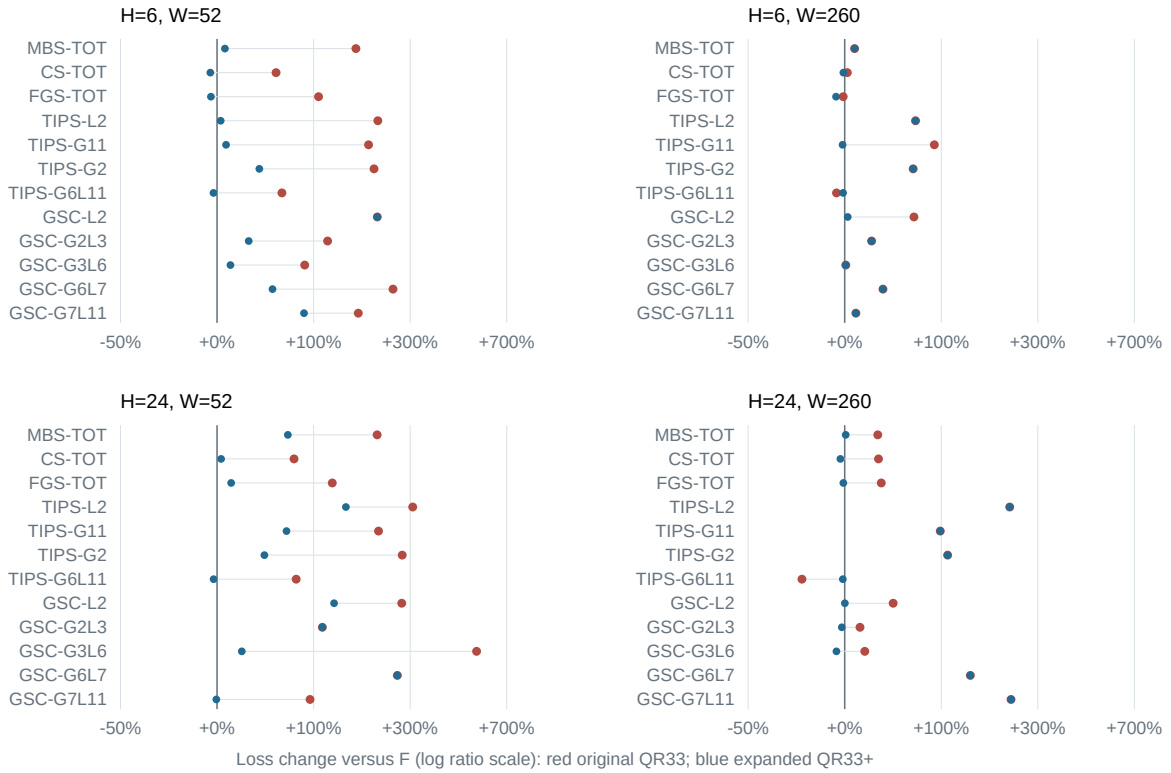


Figure 3: Original and expanded-menu regression compared with matched historical headroom in every additional weekly panel cell. Relative losses use a logarithmic ratio scale; zero marks the historical reference. Row labels omit the common series prefix.

The 24-step FX gain trades greater underprediction for less overprediction. Its ranking reverses above an underprediction/overprediction weight ratio of 20.6. For fixed reports, $J_\omega = \omega U + V$ gives this comparison, with the original score corresponding to $\omega = 19$.

Both main comparisons change ranking across consistent quantile scores. Each elementary-score curve takes positive and negative values. Appendix A.6 and Figure 5 show these crossings.

Table 8: Error components and strict breaches for the original comparisons.

Comparison	ΔU	ΔV	Crossing ω	Breaches A/B
FX six-step main comparison	-0.010	-0.921	–	376/332
FX six-step matched	-0.030	-0.776	–	376/360
FX 24-step comparison	+0.168	-3.463	20.631	531/287
Treasury main comparison	+1736.223	-4968.078	2.861	54/28

A denotes regression and B its stated historical reference; amounts are source-reported USD millions, and negative differences favor regression. A missing crossing means both components share a sign; ω represents fixed-report preferences.

Fitted quantile reports can cross across horizons even though realized peaks increase. Long-history Treasury regression crosses on about 29% of common issues. Raising a longer-horizon report to remove a crossing can worsen its score, as Appendix A.4 illustrates.

The earlier-period exercise keeps the forecast target and uses shorter weekly histories. Each weekly cell has 52 evaluation points, and the FX comparison has 1,440 origins. The FX span is about 12 calendar weeks; the weekly set includes Treasury bills and the additional panel. The earlier origins were first evaluated here, although the rules were designed after seeing the original results.

The conservative one-standard-error rule favours stronger shrinkage when validation losses are close. It selects the most shrunk candidate within one paired-loss HAC standard error of the best, including the intercept-only option. The rule adapts Hastie et al. (2009) to paired forecast losses.

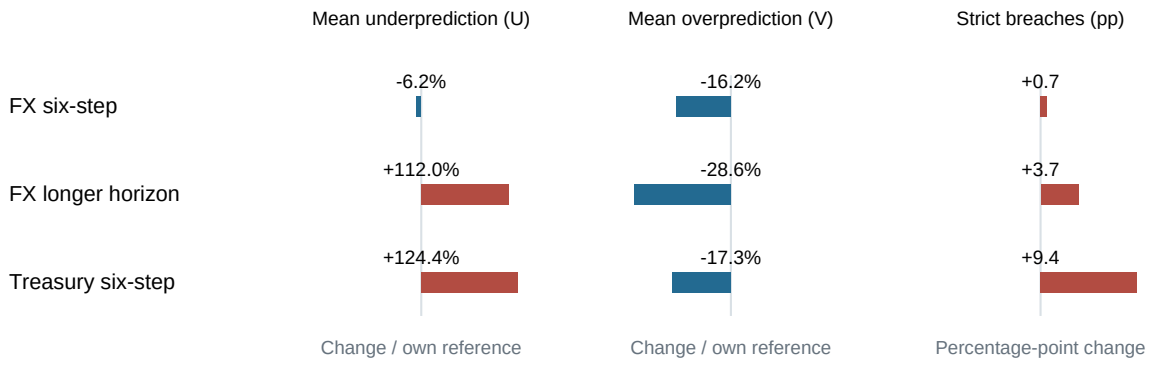


Figure 4: Changes in underprediction amounts, overprediction amounts and strict breaches for the original comparisons. Each amount panel scales its own reference to 100; breaches use percentage-point changes. The two sources retain their distinct position units.

Table 9: Robustness checks for residual replay, quantile ranks, loss weights and horizon ordering.

Check	Finding	Interpretation
AR residual replay	Lower loss than F in 5/8 and S in 5/8; more breaches than S in 8/8	Lower loss than expanded-menu regression in 3/4 Treasury cells
Rank convention	Ordinary versus corrected rank changes some finite reports; full-pool identities persist	Report construction matters
Fixed linear weights	FX short-horizon components both fall; longer-horizon crossing near the original weight	Mean score and breach count differ
Horizon order	Targets are ordered; fitted reports can cross	Separate fits need an explicit reconciliation criterion

All three pre-set success criteria fail for this conservative rule. Weekly excess loss has median +5.4% and worst +181.4%. The worst cell is Treasury bills at the 24-record horizon.

Both the protection and median-improvement requirements fail. The fixed protection limits are 25.0% for the worst cell and 5.0% for the median. Standard validation has a lower median excess loss of +4.2%, and 5 conservative-rule cells exceed the worst-cell limit.

Frequent selection of the intercept-only option coexists with large weekly losses. The conservative rule chooses it in 17 of 26 cells. The ordinary empirical quantile and corrected-rank headroom use different order statistics, as Section 4 explains.

The conservative rule also gives up more than half the eligible FX gain. Long-history loss changes by -6.3%, against -16.6% under standard validation. This misses the pre-set gain-retention criterion.

Rolling cross-validation produces two failed FX cells. A required validation candidate fails to complete in those cells. Both failures occur at the short history, one at each horizon.

The earlier FX period repeats the matched gain but reverses the rolling-reference comparison. The loss changes are -16.6% and $+3.2\%$, respectively. Table 10 pairs the same regression reports with the two historical references.

The earlier rolling-reference difference is statistically imprecise. The matched and rolling comparisons have $p = 2.94 \times 10^{-5}$ and $p = 0.574$, respectively. Table 10 identifies each reference and its inference settings.

The earlier Treasury comparison retains the adverse regression direction. The short-window version raises loss by 144.3% at $W = 52$. Table 10 uses the same weekly evaluation issues for both reports.

Earlier six-step structure	Historical reference	Origins	Δ (%)	HAC p
Treasury regression, $W = 52$	Frozen F_{52}, $d = 13$	52	+144.3	—
FX regression, $W = 2880$	Frozen F_{2880}, $d = 24$	1,440	-16.6	2.94×10^{-5}
FX regression, $W = 2880$	Rolling F_{720}, $d = 1$	1,440	+3.2	0.574

Table 10: Earlier-period six-step regression compared with matched and rolling historical rules.

Positive Δ denotes higher mean pinball loss; the weekly comparison has 52 origins and the common FX period has 1,440 (about 12 calendar weeks). HAC p-values use bandwidth 7 and are unadjusted; the Treasury short-window version at $W = 52$ differs from the original $W = 260$ comparison.

8 Implications for position monitoring

Benchmark against historical headroom. Compare conditional reports with historical headroom on the same completed labels and update schedule. Original regression loses all Treasury-bill comparisons and most additional weekly panel cells.

Default to corrected-rank headroom in short weekly samples. Use the corrected-rank historical rule as the starting point for these short samples. Wider penalties and conservative one-standard-error selection leave large losses, while the ordinary empirical quantile's excess loss reaches +36.5%.

Report breaches with the score. Show underprediction amounts, overprediction amounts and breach counts alongside pinball loss. The main FX gain accompanies more breaches, so a lower score alone misses part of the monitoring outcome.

9 Discussion and limitations

The two data sources describe different dealer-position records. The FX client-flow series samples part of one dealer's activity, while Treasury aggregates positions across reporting dealers. Both omit movements between recorded observations. Treasury observations use a downloaded vintage indexed by economic date, and aggregation can hide offsetting positions and valuation effects.

The comparisons describe forecasting performance within these records. The original FX evaluation period was inspected before the conditional methods were specified. The procedure comparisons are not causal and do not identify the effects of individual predictors. Shared reporting and overlapping positions also connect the additional weekly series.

Forecast scores and breach counts answer different monitoring questions. Their interpretation depends on the response, rank convention and chosen score. The study provides no conditional-coverage guarantee and measures no trading, collateral or funding benefit.

10 Conclusion

Simple historical rules are hard to beat when forecasting peaks of dealer net positions. In hourly FX, regression improves on matched historical headroom in both evaluation periods (14.8% and 16.6%). Its gain over the short-history rolling rule appears only in the original period; earlier loss increases 3.2%.

On weekly positions, stronger regularization narrows regression's gap, but 34 of 48 panel cells still lose. Conservative one-standard-error selection also leaves large weekly losses. Lower pinball loss and fewer breaches diverge in the main FX comparison, and both main Murphy curves cross. Position-limit model selection should therefore compare historical rules and report both error amounts and breaches. These retrospective findings from two sources provide no conditional-coverage guarantee and measure no trading or funding benefit.

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A Finite geometry and parameterization

The finite-array arguments use a common nonempty historical set J, an available rank $1 \le j \le |J|$ and real arithmetic. Write $x = N_t$, $B = |x|$, $B_i = |N_i|$, and $d_{i,h} = N_{i+h} - N_i$, including $d_{i,0} = 0$. The quantities $\ell_i = \min_h d_{i,h}$ and $u_i = \max_h d_{i,h}$ retain the notation of the main text.

A.1 Common-set projection bounds

Define $A_i = B + P_i - B_i$, $T_i = \max_h |x + d_{i,h}|$ and the floor projection $\Pi_B(v) = \max(B, v)$. Both A_i and T_i are at least B. A nondecreasing projection commutes with a finite order statistic, so $F_J = Q_j(A)$, $S_J = Q_j(T)$ and $K_J = Q_j(\Pi_B(P))$. The projection is one-Lipschitz; therefore

$$|A_i - \Pi_B(P_i)| \le |B - B_i|, \qquad |T_i - \Pi_B(P_i)| \le |x - N_i|. \tag{8}$$

The second bound follows by applying the reverse triangle inequality to every path point, then taking the maximum and applying the floor projection. Order statistics are nonexpansive in the maximum paired-row difference. If $|a_i - b_i| \le R$ for each i, then $a_i \le b_i + R$ and conversely. Thus $|Q_j(a) - Q_j(b)| \le R$. Consequently,

$$|F_J - K_J| \le R_{\text{abs}}, \quad |S_J - K_J| \le R_{\text{signed}}, \quad |F_J - S_J| \le R_{\text{abs}} + R_{\text{signed}}, \tag{9}$$

where $R_{\text{abs}} = \max_{i \in J} |B - B_i|$ and $R_{\text{signed}} = \max_{i \in J} |x - N_i|$. Among size-$m$ subsets, choosing the smallest signed distances minimizes their largest distance, since every other subset contains a distance at least as large as the mth ordered distance. Pinball loss is $\max(\tau, 1 - \tau)$-Lipschitz in its report, so the same bounds apply to absolute score differences on common rows and rank. The bounds permit either score ranking.

A.2 Envelope plateaus and signed anchors

Put $v_i = -\ell_i \ge 0$. The headroom of an increment envelope at anchor z is

$$g_i(z) = \max(z + u_i, -z + v_i) - |z| = \begin{cases} \max(u_i, v_i - 2z), & z \ge 0, \\ \max(u_i + 2z, v_i), & z \le 0. \end{cases} \tag{10}$$

The branches agree at zero. Thus g_i is two-Lipschitz and lies between $\min(u_i, v_i)$ and $\max(u_i, v_i)$. Comparing $g_i(N_i)$ with $g_i(x)$ row by row and using order-statistic stability gives

$$|F_J - S_J| \le \max_{i \in J} \min\{2|x - N_i|, |u_i - v_i|\}. \tag{11}$$

Symmetric envelopes, $u_i = v_i$, give $F_J = S_J$. Nonnegative historical and query anchors with $u_i \ge v_i$ also give equality; reversing the anchor signs interchanges the envelope sides. The same range bound yields $F_J, S_J \le B + Q_j\{\max(u_i, v_i)\}$. If every $B_i \le B$, then $F_J \ge K_J$; if every $B_i \ge B$, the order reverses. Both follow directly by comparing A_i with $\Pi_B(P_i)$.

Equal absolute anchors can still leave a translated-path gap. Take $c > 0$, $H = 1$, $t = 721$ and the history $N_s = (-1)^{s+1}c$. The completed pool $J = \{1, \dots, 720\}$ contains equal counts of both signs and the query is c. All native peaks are c and all headrooms vanish. Raw translation gives half the peaks c and half $3c$, so rank 685 yields $F = K = c$ and $S = 3c$. Here $R_{\text{abs}} = 0$ and $R_{\text{signed}} = 2c$.

A.3 Compatible futures with opposite score rankings

Fix $d > 0$ and an available corrected rank for a pool of size n. Set $t = H + n - 1$, $N_s = d(t - s)$ through issue, and use origins $J = \{0, \dots, n - 1\}$. Every historical episode declines, giving $G_i = 0$, while the current anchor is zero. Hence $F = 0$ and $S = Hd$. A future staying at zero gives peak

zero and strictly favors F; a continuation $N_{t+h} = -dh$ gives peak Hd and strictly favors S. The identical observed history therefore permits either realized score ordering.

Report order can also differ across constructions. The artificial history $(0, -4, -1, 2)$ with $H = 1$ has completed origins $0, 1, 2$ and current anchor two. Their native peaks are $(4, 4, 2)$, headrooms $(4, 0, 1)$ and translated peaks $(2, 5, 5)$. At the available corrected rank three for $\tau = 3/4$, this gives $F = 6$, $S = 5$ and $K = 4$. This auxiliary quantile belongs only to the illustration.

A.4 Fixed-coefficient residual replay

Fix arbitrary c, ϕ and define residuals $\epsilon_s = N_{s+1} - c - \phi N_s$. Replaying a historical residual sequence from x gives $M_{i,0}(x) = x$ and $M_{i,h+1} = c + \phi M_{i,h} + \epsilon_{i+h}$. Subtracting the observed recursion and using induction gives

$$M_{i,h}(x) = N_{i+h} + \phi^h (x - N_i), \qquad T^{\text{raw}}_{i,h}(x) = N_{i+h} + (x - N_i). \tag{12}$$

Taking absolute maxima and using the reverse triangle inequality gives

$$\left| \max_h |T^{\text{raw}}_{i,h}(x)| - \max_h |M_{i,h}(x)| \right| \le \max_{0 \le h \le H} |1 - \phi^h| \, |x - N_i|. \tag{13}$$

For $0 \le \phi \le 1$, the coefficient is $1 - \phi^H$; at $\phi = 1$ the paths coincide. The identity follows algebraically for any residual sequence. Scenario replay treats historical blocks as representative of future paths. The replay's time order generally requires more information than the two raw increment extrema.

Horizon order alone leaves score effects unresolved. For example, $B = 0$, both realized peaks equal one, the short report is ten, and the long report is one. Raising the long report to the short one changes its quantile loss from zero to 0.45. This separates a known response floor from an ordering imposed between fitted reports.

A.5 Peak parameterization of headroom regression

The current-magnitude predictor satisfies $B_i = \mu_{\kappa B} + \sigma_{\kappa B} z_{i,\kappa B}$ under the stated transform. Set

$$\alpha = \mu_{\kappa B} + s_\kappa \beta_0, \qquad \theta = s_\kappa \beta + \sigma_{\kappa B} e_B. \tag{14}$$

Positive homogeneity of pinball loss implies that multiplying the objective in (6) by s_κ gives

$$\mathcal{P}_\lambda(\alpha, \theta) = \frac{1}{W} \sum_{i \in \mathcal{T}} \rho_\tau(P_{i,H} - \alpha - z_{i,\kappa}^{\mathsf{T}} \theta) + \frac{\lambda}{2 s_\kappa} \|\theta - \sigma_{\kappa B} e_B\|_2^2. \tag{15}$$

The issued report is $\max\{B_t, \alpha + z_{t,\kappa}^{\mathsf{T}} \theta\}$. The equivalent peak problem thus has a shifted slope penalty. For every response $P \ge B$, raising a report below B to B weakly reduces its realized pinball loss. Translation by a known B also translates the corresponding conditional quantile.

A.6 Elementary quantile scores

For a threshold η, the elementary score for forecast c and outcome p is

$$S_{\eta,\tau}(c, p) = \{\mathbf{1}(p < c) - \tau\}\{\mathbf{1}(\eta < c) - \mathbf{1}(\eta < p)\}. \tag{16}$$

Ehm et al. (2016) represents quantile-consistent scores as nonnegative mixtures of elementary scores under the stated regularity conditions. The empirical difference is constant between forecast and outcome breakpoints, including the equality conventions at each breakpoint. Both main comparisons take positive and negative values over the full breakpoint set. Neither forecast therefore dominates its reference across all consistent quantile scores in this sample. Figure 5 displays a reduced grid with the same crossing conclusion.

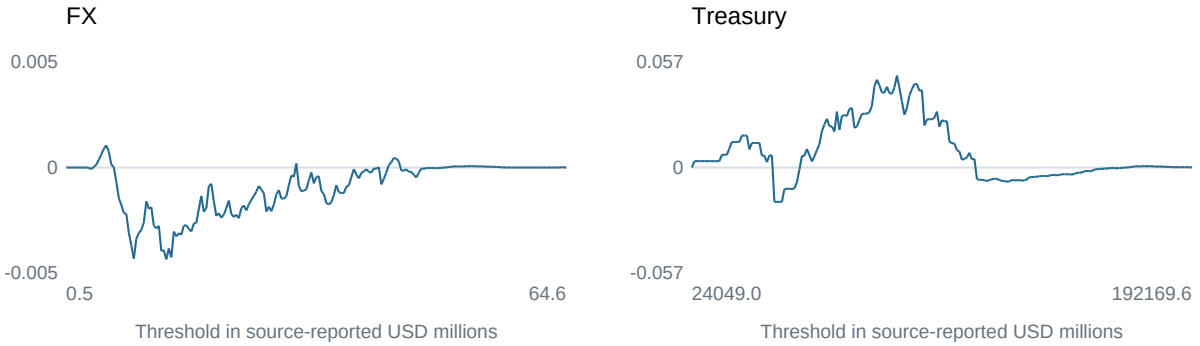


Figure 5: Elementary-score differences for the two main comparisons, with regression minus its stated historical reference. Each curve takes positive and negative values, and thresholds retain their source scales. The displayed grid preserves the crossings found over all breakpoints.

A.7 Penalty paths

The penalty paths show that greater shrinkage can raise or lower evaluation loss. Figure 6 extends the grid for both original six-step long-history settings, using fixed validation-selected models. The intercept-only optimum can be an interval of reports. Evaluation curves describe the fitted procedures; validation determines the selected penalty.

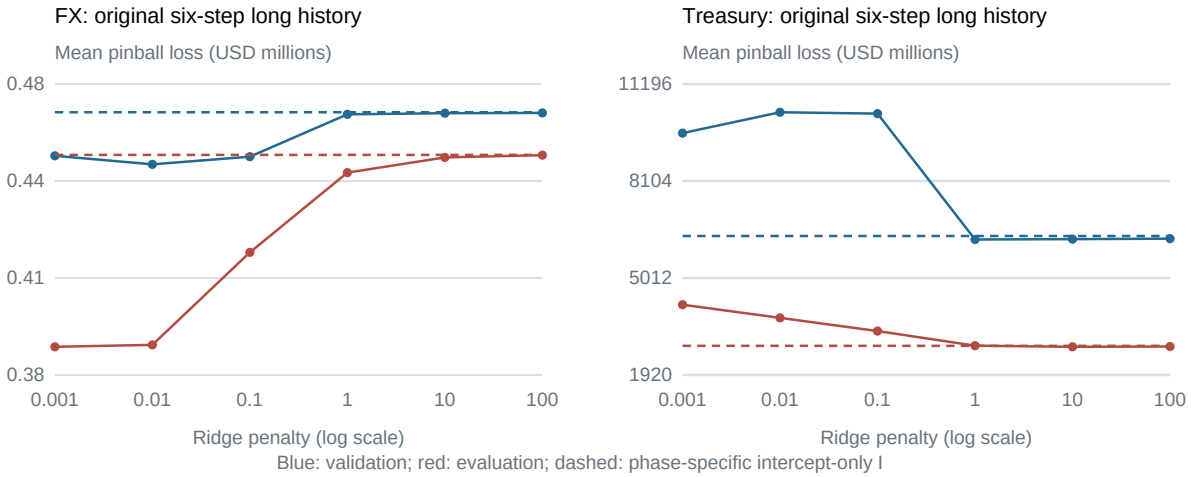


Figure 6: Validation and evaluation loss paths for the original long-history settings, with phase-specific intercept-only references. The penalty grid extends beyond the selection menu. Selected penalties use validation losses.

B Earlier construction evidence

Nearest-state retrieval improves its native-peak reference but still loses to the stronger historical headroom and translated-path rules. The earlier comparison uses recent and nearest signed-state pools on the same record. Table 11 gives the historical construction grid. Here R denotes the symmetric-increment-envelope rule and $C^{\text{floor}} = B$ the issue-floor report.

Table 11: Historical construction grid, with mean FX losses in USD millions.

Setting	F	S	R	K	C^{floor}	$F - S$
main	0.442	0.451	0.528	0.926	1.814	-0.009
h12	0.593	0.605	0.690	1.059	2.951	-0.012
h24	0.777	0.748	0.803	1.198	4.631	+0.029
n480	0.450	0.461	0.533	0.885	1.814	-0.011
n1440	0.444	0.448	0.527	0.932	1.814	-0.004
tau90	0.660	0.659	0.763	1.477	1.718	+0.000
tau99	0.155	0.159	0.175	0.289	1.890	-0.004

Each row changes one factor from the original six-step setting while retaining its evaluation origins; quantile preferences define separate tasks. The tau90 entry +0.000 is positive before rounding.

C Reproducibility and technical supplement

The repository links the original comparisons, subsequent diagnostics and document sources to their specifications. The revision guide maps every table and figure to its inputs and gives the compilation command. All stored outputs retain the complete adverse cases, numerical failures, source identities and execution context, including the longer coverage tables and loss-weight frontiers. The supplied PDF is compiled from the delivered TeX with the bundled Tectonic engine.

Exact recomputation uses the pinned source snapshots, model specification and code. The FX record belongs to the source family studied by Wisniewski et al. (2020); exact equality with their input file remains unverified. Treasury sources carry attribution and terms; FX redistribution rights remain unresolved. Public outputs contain aggregates and numerical diagnostics, while per-origin financial arrays remain private; later downloads can differ from the pinned version.

The original ridge optimizer uses two-block ADMM (Boyd et al., 2011). Accepted fits require finite calculations, the stated primal and dual residual conditions, and a feasible-dual objective-gap check. All eight expanded-menu cells and all 40 validation candidates completed, and reused four-predictor outputs were reconciled with their stored arrays and selections. The floating-point gap certificate bounds optimization error in the stated objective. An unpenalized intercept can have a flat minimizer interval: at $\tau = 19/20$, nineteen zeros and one positive value M admit every intercept in $[0, M]$.

Sequence of design and computation records.

Stage	Record
Historical construction	Historical-rule and signed-state retrieval results provide the starting comparisons.
Conditional comparison	The two main comparisons and original tuning menus precede their conditional evaluation runs.
Regression diagnostics	Common-menu specifications precede their new fits; panel predictions precede the expanded-menu panel run and follow the original panel results.
Earlier-period selection	The conservative rule, periods, success criteria and failure policy precede the earlier-period calculations, which finish before the original-period calculations start.
Development and writing	Original-period selection-rule results are used for development; rolling cross-validation retains two failed FX cells in each period, four in total.

The split between the two main comparisons and the remaining 71 matched contrasts was chosen after observing results. The former 72-comparison adjustment remains stored; the cross-window FX comparison and historical contrasts lie outside the matched family. Expanded-menu paired tests form a separate 28-comparison Holm family. The earlier-period table uses the same original regression reports for both FX references, with selected penalties $\lambda = 0.1$ for Treasury and $\lambda = 0.01$ for FX.

Applying the Harvey–Leybourne–Newbold correction to pinball loss is approximate because it originates in mean-squared-error comparisons. HAC reference distributions assume serial regularity and

a suitable finite long-run variance; the fixed-window interpretation concerns unconditional predictive ability. Nonoverlapping targets can still share dependent estimation histories, and Treasury's 40- and 12-observation grids give coarse asymptotic references. Floor atoms distinguish the exact nominal-exceedance null from the upper-bound condition for a quantile. POF uses all retained hits, while independence and joint coverage use likelihoods consistently conditioned on the first hit; the earlier mixed-initialization calculation is preserved separately.

Numerical details relocated from the main text.

Comparison	Recorded detail
Original six-step FX HAC	Approximately -5.2 against rolling headroom; -5.9 against matched frozen headroom.
Ordinary intercept-only Treasury quantile	Loss changes of $-2.0\%, +3.2\%, +6.2\%, +2.4\%$ relative to F, in Table 6's Treasury row order.
Expanded-menu Treasury regression	The upper menu endpoint is selected in 3 of 4 cells.
Original Treasury main inference	Unadjusted HAC $p = 0.008$; two-comparison Bonferroni $p = 0.015$.
Treasury selected penalty	The main setting changes from $\lambda = 0.001$ under original regression to $\lambda = 1$ under expanded-menu regression.

Internal labels retained in stored outputs and figure images, with reader-facing names and fixed prediction outcomes.

Stored label	Description or recorded outcome
R0	Standard validation.
R1	Validation with the intercept-only option.
R2	The conservative one-standard-error rule.
R3	Rolling cross-validation.
S1 / S2 / S3	The three pre-set success criteria: weekly protection, FX gain retention and weekly median improvement; all fail.
E3	The common-menu regression comparison.
E5	The additional weekly panel with an expanded penalty menu.
E6	The extended penalty paths.
R6	The earlier-period selection-rule comparison and original-period development results.
33+ / Four+ / Original 33	Expanded-menu regression / four-predictor regression / original regression.
P1	Improvement over original regression in originally losing panel cells: 30/45 against a threshold of 40; unsupported.
P2	Panel losses within the pre-set band around F: 24/48 against a threshold of 36; unsupported.
P3	Panel penalties $\lambda \geq 1$: 33/48 against a threshold of 25; supported.

C.1 Notation

Notation used in the comparison and appendices.

$N_t; B_t; P_{t,H}; G_{t,H}$	Signed position; issue magnitude; peak; headroom
$H; W; d; \kappa(t)$	Record horizon; completed-label count; refit interval; latest refit
$s_\kappa; z_{i,\kappa}; \lambda$	Response scale; standardized predictors; ridge penalty
$\ell_i; u_i; T_{i,h}^{\text{raw}}$	Lower/upper path increments; raw translated path
$R_{\text{abs}}; R_{\text{signed}}$	Maximum absolute-anchor/signed-anchor distances
$U; V; \omega$	Mean underprediction; mean overprediction; their weight ratio
$C^{\text{floor}}; \mathcal{P}_\lambda$	Issue-floor report; penalized training objective